

Technical Datasheet — VBF-T1R1-DSH-01

Case: t1_r1 | Date: 2026-08-25 | Prepared: _____ | Reviewed: _____ | Approved: _____

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Rated capacity	5.0 Ah nominal (parameter); 5.6505 Ah simulated 1C discharge	Chen2020 "Nominal cell capacity [A.h]"; cell/r3_I_1c.json:capacity_ah
Nominal voltage	4.0067 V (discharge midpoint)	cell/r3_I_energy.json:midpoint_voltage_v
Voltage window	2.5 – 4.2 V	Chen2020 cut-off parameters
Rated energy	19.967 Wh	cell/r3_I_energy.json:energy_wh (V·I time integration)
Energy density	553.98 Wh/kg; 905.98 Wh/L	cell/r3_I_energy.json:energy_density_wh_kg / energy_density_wh_l (contract mass caliber: layer stack, electrolyte excluded)
Maximum continuous discharge	1C (5.0 A); verified 5.6505 Ah at 1C, 298.15 K ambient, peak temperature 299.55 K	cell/r3_I_1c.json (protocol 1C_discharge)
Fast-charge capability	4C (20 A) CC charge: peak cell temperature 325.65 K (52.51 °C, ambient 318.15 K, rise +7.50 K); no lithium plating (anode potential min +0.0249 V)	cell/r3_I_4c_dfn.json (DFN authoritative; SPMe corroborates min +0.0135 V)
DC internal resistance	0.155 m Ω ? (mechanical derivation: $(V[0] - V[10\% t]) / I_{1C}$)	cell/r3_I_energy.json:dcr_ohm
Peak power density	748.3 kW/kg ($P = V_{OC2}/(4 \cdot R_{DC})$, per mass)	cell/r3_I_energy.json:power_density_w_kg
Operating temperature range	298.15 K discharge / 318.15 K 4C charge simulated; wider range not simulated (honest statement)	protocol definitions (1C_discharge, 4C_charge_45C)
Cycle life	Not simulated (requires aging model) — must not be fabricated	—
Safety determination	Overcharge to 4.7000 V: no thermal runaway (triggered = false, peak 299.55 K); 4C charge below 60 °C red line	cell/r3_I_oc.json; cell/r3_I_tr.json; cell/r3_I_4c_dfn.json
Dimensions	Electrode sheet 65 mm × 1580 mm; stack thickness 214.6 μ m; shell dimensions Not provided	Chen2020 geometry; cell/r3_I_energy.json:thickness_m
Mass	36.044 g stack (contract caliber, electrolyte excluded); 45.43 g with electrolyte fill (1.2 g/cm ³ literature density)	cell/r3_I_energy.json:mass_kg; pore-volume derivation (see design_spec section 3)
Chemistry	NMC811 / graphite; high-conductivity high-t+ electrolyte ($\sigma = 1.8$ S/m estimate, t+ = 0.55 estimate)	Chen2020 + params_r3_I.json